Linear Algebra (MT-1004)

Homework#08

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Question 1

Let

$$A = \begin{bmatrix} 4 & 1 & -2 \\ 1 & 3 & 0 \\ -2 & 0 & 3 \end{bmatrix}$$

be a 3×3 real matrix.

(a) Find the characteristic polynomial of the matrix A.

(b) Compute all eigenvalues of A.

(c) For each eigenvalue, determine the corresponding eigenvectors.

(d) Verify that the eigenvectors corresponding to distinct eigenvalues are linearly independent.